

Assignment 2: Data Cleaning and Transformation

Payments/Retail Product Dataset — Cleaning Process, Screenshots & Explanations

1. Handling Missing Values

(Screenshot — Raw Data (missing Price / Category cells shaded in red))

1.a Missing Price:

Missing Price Count – 3 rows had a blank price

Sony Headphones – Category: Electronics

Coleman Camping Tent – Category: Outdoor

Ray-Ban Glasses – Category: Fashion

Strategy: Missing prices were imputed using Median price of other products of the same category. Rows were kept without deleting it quantity

1. b Missing Category:

Missing Category Count – 4 rows had a blank price

North Face Backpack – No match found in the dataset, imputed as accessories

Adidas Sneakers – Matched Category, imputed as Fashion

Nespresso Coffee Maker – Matched Category, imputed as Kitchen

Xiaomi Fitness Tracker – Matched Category, imputed as Electronics.

Strategy: Missing Category were imputed by matching the product name to the Category that was already used for the same product.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	Laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	Smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	
07-JUN-UK	Headphones	Sony		45	Electronics
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	
08-DEC-DE	Coffee Maker	Nespresso	120	35	
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
16-APR-ES	Headphones	Bose	250	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman		10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
29-SEP-CA	Smartphone	Google	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion
11-JUL-CA	Blender	Vitamix	400	40	Kitchen
16-APR-ES	Headphones	Bose	250	20	Electronics
07-MAR-CA	Dress	Zara	60	30	Fashion
13-APR-CA	Toaster	Hamilton	40	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics
02-DEC-CA	Jeans	Levi's	50	50	Fashion
17-JUN-IN	Laptop	HP	950	25	Electronics
09-JUL-FR	Watch	Casio	100	20	Accessories

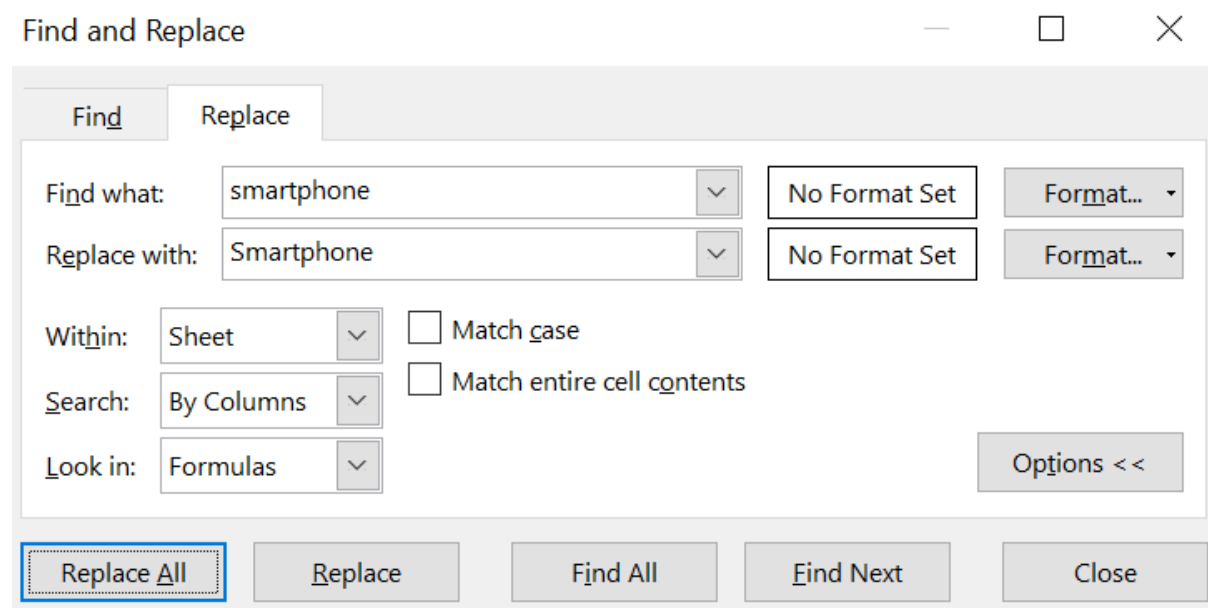
2. Correcting Inconsistent Data

Product Name Inconsistency – Some names were in lowercase (laptop, smartphone, headphones) while others were properly capitalized, (Sneakers, Coffee Maker)

Fix Applied – Used Find & Replace to convert all product name entries to keep it consistent
eg: laptop – Laptop

Category Typos: Several rows had “Electroni” instead of “Electronics” (Headphones/Sony, Fitbit/Smartwatches)

Fix Applied – Find & Replace “Electroni” – “Electronics”



Find and Replace

Find Replace

Find what: smartphone No Format Set Format...

Replace with: Smartphone No Format Set Format...

Within: Sheet Match case

Search: By Columns Match entire cell contents

Look in: Formulas Options <<

Replace All Replace Find All Find Next Close

3. Removing Duplicates

Method – Compared every column (Product ID, Product Name, Brand Name, Price, Quantity, Category) for each row: rows fully identical to an earlier row were removed.

Samsonite Laptop Bag – Product ID (21-AUG-CA)

Bose Headphones – Product ID (16-APR-ES)

HP Laptop- Product ID (17-JUN-IN)

Result – 3 Duplicates rows were removed, Hence Dataset reduced from 34 – 31 rows.

4. Splitting and Merging Data

Product ID Split – Product ID (eg – “28-JAN-US”) was split on the “-” delimiter into Manufacturing Date (day-month) and Country (last segment)

– Since source product ID has no year component. A placeholder year of (2026) current year was assumed for all manufacturing date values so the DD-MM-YYYY format has could be applied. Corrected confirming the manufacturing year

Brand + Product Name merged = Brand Name and product name were concatenated merged into a new Product Brand. Eg.. Dell + Laptop = Dell Laptop

5. Number Formatting

Price Column – Formatted as currency: eg. \$1000.00

Manufacturing Date Column – Formatted Date as DD-MM-YYYY eg.. 28.01.2026

6. Conditional Formatting

Price Column – Applied a 3-color scale (red=lowest, yellow=mid, green = highest) so price ranges are visible easily with a glance

Category Column - Applied custom rule that highlights any cell with “Electronics” in green bold dark green text.